

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 6

Total Marks:30

Annexure A

Scenario-Based

Code of Conduct:

*I Deepa lakshana.B(192471054) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Scenario-Based

1. Use Case Diagram Scenario

Actors

- Customer
- Pharmacist
- Administrator

Use Cases

Customer

- Register/Login
- Search Medicines
- Upload Prescription
- Place Medicine Order
- Make Online Payment
- Track Order

Pharmacist

- Verify Prescription
- Dispense Medicines
- Update Order Status

Administrator

- Manage Medicine Inventory
- Manage User Accounts
- Generate Reports

Relationships

- <<include>>
 - Place Order → Search Medicines
 - Place Order → Make Payment
 - Dispense Medicine → Verify Prescription
- <<extend>>
 - Upload Prescription extends Place Order (only for prescription medicines)
- **Generalization**
 - User
 - Customer
 - Pharmacist
 - Administrator



Customer:

Login

Search Medicine

Upload Prescription

Place Order

Make Payment

Track Order

Pharmacist:
Verify Prescription
Dispense Medicine
Update Order

Administrator:
Manage Inventory
Manage Users
Generate Reports

Place Order
 <<include>>
Search Medicine
Make Payment

Upload Prescription
 <<extend>>
Place Order

Dispense Medicine
 <<include>>
Verify Prescription

2. Class Diagram Scenario

Classes

Guest

Attributes

- guestID
- name
- phone

Methods

- bookRoom()
 - cancelReservation()
-

Room

Attributes

- roomNo
- type
- status

Methods

- checkAvailability()
 - updateStatus()
-

Reservation

Attributes

- reservationID
- checkIn
- checkOut

Methods

- createReservation()
 - cancelReservation()
-

Payment

Attributes

- paymentID
- amount
- paymentMode

Methods

- processPayment()
-

Receptionist

Attributes

- employeeID
- name

Methods

- assignRoom()
 - checkInGuest()
-

Relationships

- Guest **1..*** Reservation (Association)
- Reservation **1..1** Room (Association)
- Reservation **1..1** Payment (Composition)
- Hotel **1..*** Room (Aggregation)
- Employee → Receptionist (Inheritance)

3. Sequence Diagram Scenario

Objects

- Customer
- Login Page
- Authentication Service
- Banking Server
- Database

Sequence

Customer → Login Page : Enter Username & Password

Login Page → Authentication Service : Verify Login

Authentication Service → Database : Validate Credentials

Database → Authentication Service : Success

Authentication Service → Login Page : Login Successful

Customer → Banking Server : Transfer Funds

Banking Server → Database : Check Balance

Database → Banking Server : Balance Available

Banking Server → Customer : Send OTP

Customer → Banking Server : Enter OTP

Banking Server → Database : Update Transaction

Database → Banking Server : Transaction Success

Banking Server → Customer : Transaction Confirmation

4. Activity Diagram Scenario

Start



Submit Application



Upload Documents



Identity Verification



Decision

Documents Valid?

YES ↓

Pay Processing Fee



Application Verification



Decision

Approved?

YES ↓

Print Passport



Dispatch Passport



End

NO

Reject Application



End

Parallel Activities (Fork)

Document Verification

Identity Verification

Join

Continue Processing

Decision Nodes

- Documents Valid?
- Application Approved?

Fork

- Document Verification
- Identity Verification

Join

- Both processes complete before approval.

5. State Diagram Scenario

States

- Idle
- Door Open
- Door Closed
- Moving Up
- Moving Down
- Emergency Stop
- Maintenance Mode

Start

↓

Idle

↓

Floor Button Pressed

↓

Door Open

↓

Door Closed

↓

Moving Up / Moving Down

↓

Destination Reached

↓

Door Open

↓

Idle

Emergency Button

↓

Emergency Stop

↓

Reset

↓

Idle

Maintenance Switch

↓

Maintenance Mode



Resume Service



Idle

6. Deployment Diagram Scenario

Hardware Nodes

- **Mobile Phone**
- **Internet**
- **Home Gateway**
- **IoT Controller**
- **Smart Sensors**
- **Cloud Server**

Software Artifacts

- **Mobile App**
- **Home Automation Software**
- **IoT Firmware**
- **Cloud Database**
- **Monitoring Service**

Deployment Architecture

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| **Mobile Phone** |

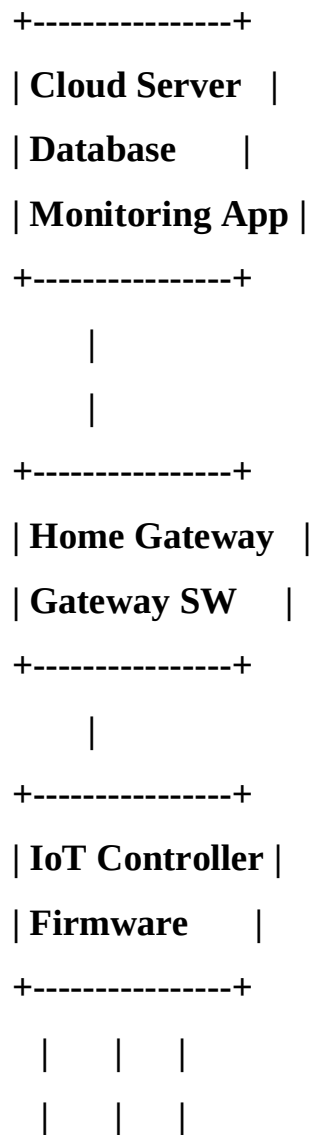
| **Mobile App** |

+-----+

|

Internet

|



Sensor Camera Smart Light

Communication Links

- **Mobile App ↔ Cloud Server**
- **Cloud Server ↔ Home Gateway**
- **Home Gateway ↔ IoT Controller**
- **IoT Controller ↔ Smart Sensors and Appliances**

RUBRICS FOR CALCULATION

Scenario-Based

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand